Mathematical Statistics | **Exam 2019-10-28 at 14:00–19:00**
Centre for Mathematical Sciences | **MASA01 A**
Lund University | **Probability theory**

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Only statistical tables provided by the institute are allowed. Electronic calculators are allowed. No other tables or formula sheets are allowed.

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1. Let $\Omega = \{1,2,3,4,5,6\}$, $\mathcal{F} = \{\emptyset, \Omega, \{1\}, \{2,3,4,5,6\}\}$, and let P be a function defined on $\mathcal{G} = \mathcal{P}(\Omega)$ the set of all subsets of Ω, such that

$$P(A) = \frac{\text{the number of elements in } A}{6},$$

for every $A \in \mathcal{G}$. Let X be a function on Ω, defined by

$$X(\omega) \quad = \quad \begin{cases} 0 & \text{if } \omega \in \{1,2\}, \\ 1 & \text{if } \omega \in \{3,4,5,6\}. \end{cases}$$

   (a) Modify the set $\mathcal{F}$ so that X becomes a random variable on the probability space $(\Omega, \mathcal{F}, P)$. [3p]

   (b) Find the probability mass function for the random variable X. [7p]

2. Suppose that X and Y are independent random variables, X is uniformly distributed on the interval $[4,6]$ and Y has the exponential distribution with expectation 1.

   (a) Calculate $P(X < Y)$. [5p]

   (b) Calculate $P(5 \leq X + Y \leq 7)$. [5p]

3. Students at a university biology department are interested in studying migrating birds at a remote island in the Atlantic ocean. For this reason the students set up nets on the island, in which the birds may be caught. If a bird is caught it is tagged (marked) with a ring on it's left leg, and let loose. Some of the birds may be caught again, some birds may be caught two times, or three times, and some birds are never caught. The probability that a bird is never caught is 0.5, that it is caught only once is 0.35, that it is caught only twice is 0.1, and that it is caught three times is 0.05. No bird is caught more than three times. The probability that a bird contracts a (harmless) virus from humans is 0.01 (for birds never caught), 0.1 (for birds caught only once), 0.2 (for birds caught only twice) and 0.3 (for birds caught three times).

(a) What is the probability that a bird contracts the virus? [4p]

(b) What is the proportion of birds that have been caught at least once, among those that contracted the virus? [6p]

4. Let X be a random variable with density

$$f(x) = \begin{cases} \frac{1}{4}\binom{5}{x}0.3^x(1-0.3)^{5-x}, & x = 0, 1, \ldots, 5, \\ \frac{3}{12}e^{x/3}, & x < 0, \end{cases}$$

with respect to the measure $\mu(x) = \mu_0(x)1\{x \geq 0\} + \mu_1(x)1\{x < 0\}$, where μ_0 is counting measure (of the integers), and μ_1 is length measure.

(a) Calculate $P(X \geq 0)$ and $E(X)$. [10p]

(b) Calculate $P(-1 < X \leq 3 \,|\, -1 < X \leq 2)$ and $E(X|X \geq 0)$. [10p]

5. The supreme leader of a big country, Duck Plopp, does not like his advisors. He would like to fire every one of them but is restrained by his chief of staff, who he can not fire. He tries to fire one advisor every month. Each month he picks one advisor at random from the ones that are still not fired, and tries to fire that advisor. The probability that he succeeds to fire the chosen advisor is 0.25. The attempts to fire the advisors at different months are independent attempts, and he does not hire new advisors when one is fired.

Duck Plopp has currently five (5) advisors, not counting his chief of staff. He has exactly one year left before he has to get a new government with new advisors.

(a) What is the probability that Duck Plopp will fire the last advisor, at exactly the last month of the year that is left? [10p]

(b) His unofficial and secret advisor advices him, that if he manages to have only the last three months without advisors he can make 5 major decisions, if he manages to have only the last two months without advisors he can make 2 major decisions, and if he has only the last month without advisor he can make 1 major decision. What is the expected number of decisions that Duck Plopp can make with the secret advisor's plan? [10p]

6. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent identically distributed continuous random variables, with distribution function F. The empirical survival function is defined by

$$S_n(t) = \frac{1}{n}\sum_{i=1}^{n} 1\{X_i > t\},$$

for any fixed real number t.

(a) Calculate $E(S_n(t))$ and $Var(S_n(t))$, for fixed t. [5p]

(b) Find the distribution of $nS_n(t)$, for fixed t. (You may either state the name of the distribution, or give a formula for $nS_n(t)$'s density or distribution function.) [7p]

(c) Calculate the covariance of $S_n(t)$ and $S_n(s)$, for fixed s and t. [8p]